

AI Incident Report

Generated: 28-07-2026 20:10:28

****CYBERSECURITY INCIDENT REPORT****

****Incident ID:**** INC-2026-001

****Report Date:**** 2026-07-14

****Prepared By:**** Senior SOC Analyst

1. Executive Summary

On July 14, 2026, at 10:30:00, the Security Operations Center (SOC) identified a high-severity brute-force attack targeting a core Ubuntu Server (10.0.0.5) within our infrastructure. The attacker, originating from IP address 192.168.1.100, attempted to gain unauthorized administrative access by systematically testing login credentials against the SSH service. The automated security controls successfully identified the malicious activity and blocked the source IP, preventing a breach of sensitive data. The incident has been fully contained and resolved.

2. Incident Timeline

- * 10:30:00 – Initial alert generated by the Intrusion Detection System (IDS) indicating an anomalous volume of failed SSH login attempts.
- * 10:32:00 – SOC analysis confirmed an active brute-force campaign originating from 192.168.1.100.
- * 10:35:00 – Automated firewall rules triggered, successfully blacklisting the source IP.
- * 10:40:00 – Server logs reviewed; verified that no successful logins were achieved during the attack duration.
- * 10:45:00 – Incident declared contained and resolved.

3. Root Cause Analysis

The incident was a targeted attempt to exploit common SSH vulnerabilities. The attacker utilized automated scripts to conduct a dictionary-based password attack against the root user account. The root cause of the attempt is the exposure of the SSH service to the internal network, combined with a lack of rate-limiting or account lockout policies on the target host, which allowed the attacker to persist in their efforts until the firewall threshold was met.

4. Evidence Collected

- * Authentication Logs: 150 failed SSH login attempts recorded in the /var/log/auth.log file.
- * Behavioral Evidence: Sustained sequence of login requests occurring within a narrow 5-minute window.
- * Administrative Target: Documentation of multiple failed attempts to access the "root" account, indicating a goal of full system compromise.
- * Network Evidence: Firewall logs confirming the successful drop of all traffic from 192.168.1.100.

5. Business Impact

The potential impact of this attack was high, as a successful compromise of the root account would have granted the attacker complete control over the Ubuntu Server, potentially leading to unauthorized data access, system modification, or lateral movement within the network. However, due to effective defensive configurations, there was no actual compromise, no data exfiltration, and no service downtime experienced by the business.

6. MITRE ATT&CK; Mapping

- * T1110.001 – Brute Force: Password Guessing (The attacker systematically guessed passwords to gain access).
- * T1078.003 – Valid Accounts: Local Accounts (The attacker specifically targeted the root administrative account).

7. Recommendations

- * Disable Root Login: Modify the sshd_config file to set 'PermitRootLogin no' to prevent direct root access.
- * Implement Key-Based Authentication: Transition from password-based SSH authentication to SSH keys and disable password authentication entirely.
- * Deploy Fail2Ban: Install and configure Fail2Ban to automatically ban IPs after a set number of failed login attempts, adding a secondary layer of protection.
- * Change Default Port: Move the SSH service from the default Port 22 to a non-standard port to reduce exposure to automated scanners.
- * Enhance Monitoring: Configure real-time alerts for failed login attempts to be sent directly to the SOC dashboard for faster response times.

8. Conclusion

The incident was a standard automated brute-force attempt that was successfully mitigated by our perimeter security defenses. While the system remained secure, this incident highlights the need for a "defense-in-depth" approach. By implementing the recommended hardening steps, we will significantly increase the security posture of the Ubuntu Server and minimize the risk of future credential-based attacks. The incident is considered closed.